



Risk of herbivore attack and heritability of ontogenetic trajectories in plant defense

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Abstract

Ontogeny has been identified as a main source of variation in the expression of plant phenotypes. However, there is limited information on the mechanisms behind the evolution of ontogenetic trajectories in plant defense. We explored if risk of attack, herbivore damage, heritability, and phenotypic plasticity can promote or constrain the evolutionary potential of ontogenetic trajectories in three defensive traits. We exposed 20 genotypes of *Turnera velutina* to contrasting environments (shadehouse and field plots), and measured the cyanogenic potential, trichome density, and sugar content in extrafloral nectar in seedlings, juveniles and reproductive plants. We also assessed risk of attack through oviposition preferences, and quantified herbivore damage in the field. We estimated genetic variance, broad sense heritability, and evolvability of the defensive traits at each ontogenetic stage, and of the ontogenetic trajectories themselves. For plants growing in the shadehouse, we found genetic variation and broad sense heritability for cyanogenic potential in seedlings, and for trichome density at all ontogenetic stages. Genetic variation and heritability of ontogenetic trajectories was detected for trichome density only. These genetic pre-requisites for evolution, however, were not detected in the field, suggesting that environmental variation and phenotypic plastic responses mask any heritable variation. Finally, ontogenetic trajectories were found to be plastic, differing between shadehouse and field conditions for the same genetic families. Overall, we provide support for the idea that changes in herbivore pressure can be a mechanism behind the evolution of ontogenetic trajectories. This evolutionary potential, however, can be constrained by phenotypic plasticity expressed in heterogeneous environments.

Keywords Defense · Genetic variation · Broad-sense heritability · Ontogenetic trajectories · Reaction norms

Introduction

Ontogeny has been recognized as one of the main sources of phenotypic variation in both plants and animals, as phenotypes drastically change throughout development (Pigliucci and Schlichting 1995; Sultan 2015). This variation results in

ontogenetic trajectories of the phenotype, defined as functions of phenotypic changes resulting from internal factors (genetic and developmental) and external environmental effects (physical conditions and biotic interactions). Hence, ontogenetic trajectories can be considered as a reaction norm of the genotype exposed to different environments across development (Schlichting and Pigliucci 1998; Schlichting and Smith 2002; Whitman and Agrawal 2009). Knowledge on the phenotypic and genetic aspects of this ontogenetic variation is relevant for our understanding of phenotypic evolution, and could reveal why some biotic interactions shift over time (Kirkpatrick and Lofsvold 1992; Larsson et al. 1998).

Ontogenetic trajectories in anti-herbivore defensive traits have now been described for many plant species (Boege and Marquis 2005; Barton and Koricheva 2010), and are considered a ubiquitous characteristic of plant phenotypes (Barton and Boege 2017). The direction and degree of such ontogenetic changes seem to be quite variable, although

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patterns have emerged with respect to plant life histories, and the specific developmental window or defensive trait examined (Barton and Koricheva 2010; Quintero et al. 2013). The emergence of some general patterns suggests that these ontogenetic trajectories constitute common functional adjustments to changes in resource allocation (Bryant et al. 1991), regulatory mechanisms (Campos et al. 2016), gene expression (Mauricio 2005), and/or developmental constraints (Villamil et al. 2013). Additional mechanisms driving ontogenetic trajectories could be due to the multifunctionality of traits (Barton and Boege 2017) or the variation in the selective pressures by consumers acting on plants throughout their lifetimes (Boege and Marquis 2005; Barton and Koricheva 2010). Despite the abundant evidence of ontogenetic changes in plant defense, there is currently not enough data on the mechanisms and sources of variation driving the evolution of these ontogenetic trajectories, necessary to assess whether they can be considered adaptive complex phenotypes under natural selection (Barton and Boege 2017).

The capacity of a population to undergo evolutionary changes in their ontogenetic trajectories requires a significant degree of evolvability, which is a measure of its potential to respond as a complex trait to natural selection and other evolutionary processes (Flatt 2005). One relevant factor that can influence this potential is the degree of genetic variation in plant defenses at each ontogenetic stage. In particular, the risk of herbivore attack, the cost of plant defense, and the fitness consequences of damage have been proposed as factors that can select for optimal defense allocation patterns (Coley 1983; Bryant et al. 1991). Because the probability of being damaged by herbivores (Bryant et al. 1992;

Hanley et al. 1995; Fenner et al. 1999; Maron and Crone 2006; Kant et al. 2015; Schuman and Baldwin 2016), the costs of defenses (Boege et al. 2007; Orians et al. 2010), and the impacts of damage (Boege 2005; Barton 2014; Tito et al. 2016) can change across plant development, natural selection should optimize the expression of defenses at each ontogenetic stage (Barton and Boege 2017). Following the optimal defense theory (McKey 1974; Rhoades 1979), vulnerable ontogenetic stages should be exposed to stronger selection pressures to increase their defenses against herbivores than developmental stages with lower risk of attack and/or less susceptible to herbivore damage. As a consequence, genetic variation should be differentially eroded by natural selection throughout ontogeny. In other words, when risk of herbivore attack is high, natural selection should favor those genotypes maximizing the investment in defenses, hence reducing the proportion of heritable variation in these traits. For example, when young stages have greater risk of attack, natural selection should favor genotypes with high levels of defense at this stage (Fig. 1a). In a scenario where risk of attack is greater at older ontogenetic stages, the opposite trend should be expected (Fig. 1b). Thus, genetic variation should be reduced at those ontogenetic stages under stronger selective pressure on plant defense. If the risk of herbivore damage is evenly distributed or unpredictable across plant ontogeny, genetic variation in plant defense should be expected to be equivalent among ontogenetic stages (Fig. 1c). Under this scenario, selective agents other than herbivory, such as resource allocation trade-offs due to other functional priorities, developmental constraints or phenotypic plasticity could promote variable ontogenetic patterns in plant defense.

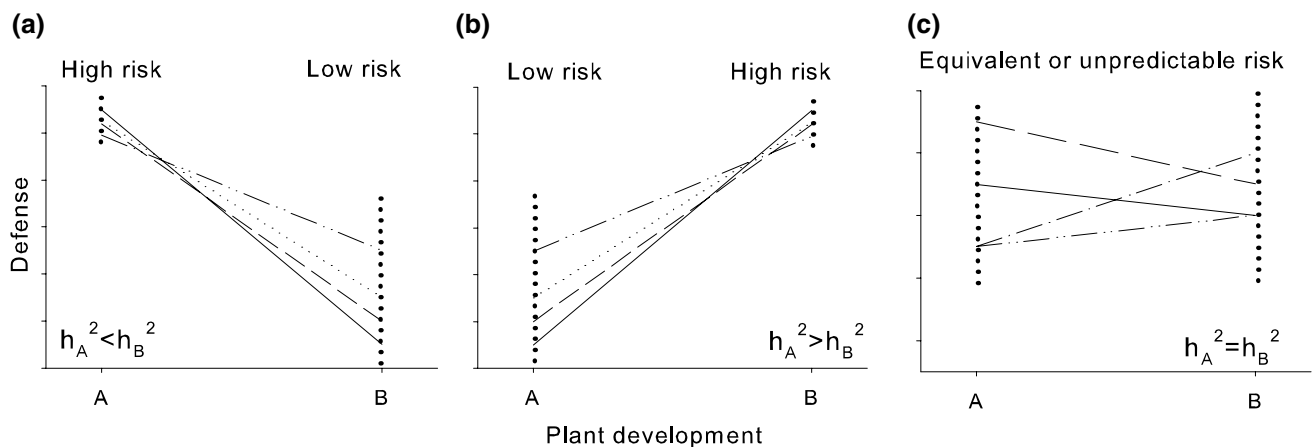


Fig. 1 Expected relationship between risk of attack and genetic variation (represented by vertical dashed lines) of plant defense across plant development. Different ontogenetic trajectories represent trends for different genotypes. **a** Younger stages have higher risk of attack, and hence natural selection favors higher levels of defense and erodes genetic variation in contrast with older stages. **b** Risk of

attack is greater in older ontogenetic stages, promoting higher defense with low genetic variation than in earlier ontogenetic stages. **c** Risk of attack is equivalent or unpredictable across plant development, hence genetic variation in plant defense remains high and similar for all ontogenetic stages, producing different ontogenetic patterns in plant defense among genotypes

The expression and evolution of ontogenetic trajectories can indeed be influenced by phenotypic plasticity at one or more ontogenetic stages (Pigliucci and Schlichting 1995). The release and/or amplification of phenotypic variation as a result of plasticity could mask heritability of plant defenses, and as a consequence of their ontogenetic trajectories. Acknowledging that both internal factors and external environments often change across plant development, one major but unanswered question is whether ontogenetic trajectories in plant defense are fixed or actually vary as a function of the environmental context in which plants develop. In other words, are the ontogenetic trajectories themselves plastic across variable environments?

A fair number of studies on single ontogenetic stages have confirmed that both constitutive and induced defenses have significant levels of heritable variation (Underwood et al. 2000; Andrew et al. 2005; Stevens and Lindroth 2005), and that the correlation among multiple traits can influence their joint evolution (Berenbaum et al. 1986; Stowe 1998). To date, what we know for a limited number of species is that genetic variation of plant defense can change across ontogeny (i.e., genotype $\times$ age interactions). For example, in *Plantago major* and *P. lanceolata*, genetic variation was detected in the expression of iridoid glycosides throughout ontogeny (Bowers and Stamp 1993; Barton 2007). In addition, there is one report on the ontogenetic changes in heritability of trichome density (Mauricio 2005), but no information is available on the heritability of the ontogenetic trajectories themselves nor their extent of phenotypic plasticity. Hence, a first step to explore the evolutionary potential of ontogenetic trajectories is the assessment of how heritable genetic variation in plant defense changes across plant development. A second step is evaluating the evolutionary potential of the ontogenetic trajectories themselves. In this study, we: (1) assessed the genetic variation and broad sense heritability of three defensive traits at three ontogenetic stages, (2) explored the relationship between heritable variation and the risk of herbivore attack and proportion of leaf area consumed by herbivores throughout plant development, and (3) estimated if heritability of the ontogenetic trajectories in defensive traits are contingent upon the environment in which plants grow. We predicted that if ontogenetic trajectories in plant defense represent a functional adjustment tuned by the selective pressure of herbivore damage, a negative relationship between levels of defense and genetic variation should be found across plant development (Fig. 1a, b). We also predicted changes in the expression of defensive traits and their genetic variation in different environments, which would in turn promote plasticity of their ontogenetic trajectories.

Materials and methods

Study system

Turnera velutina (Passifloraceae) is a myrmecophytic shrub endemic to Mexico growing in coastal sand dunes and tropical dry forests (Arbo 2005; Villamil et al. 2013). Because we were particularly interested in assessing the evolutionary potential of the ontogenetic trajectories in plant defense within a population under field and greenhouse conditions, we focused on one population of *T. velutina* located on the coast of Veracruz, Mexico (19°36' N, 96°22' W), within the Centro de Investigaciones Costeras La Mancha (CICOLMA). The climate in this area is warm subhumid with an average annual temperature ranging between 21.1 °C in January and 27.3 °C in June. The rainy season occurs mostly from June to September, and the total annual precipitation ranges from 899 to 1829 mm (Travieso-Bello and Campos 2006). Ontogenetic trajectories in chemical (cyanogenic potential), physical (trichomes density), and biotic (sugar in extrafloral nectar as a reward for patrolling ants) defensive traits have been previously reported for *T. velutina* as follows: whereas the cyanogenic potential (HCN) decreases from the seedling to the reproductive stage, both trichome density and sugar in extrafloral nectar (EFN) increase across plant development (Ochoa-López et al. 2015). In addition, previous reports indicate that younger ontogenetic stages have greater percentages of consumed leaf area than later stages (Ochoa-López et al. 2015). The main herbivore of *T. velutina* is the specialist herbivore *Euptoieta hegesia* Cramer (Lepidoptera: Nymphalidae), which is commonly found in the studied population (Ochoa-López et al. 2015).

To reduce maternal effects, in June 2013, we produced a F1 generation ($N = 300$ plants) of *T. velutina*, using seeds from 20 maternal plants naturally growing in the field in different sites of the dry forest vegetation surrounding the established sand dunes. To obtain the seeds, multiple flowers of each plant were self-crossed and further excluded from pollinators. Fruits were collected when ripe. After removing their elaiosomes, seeds were sown in germination trays in a mixture of local soil and vermiculite (1:1). Trays were bottom watered for 3 weeks until germination. F1 plants were transplanted into 2-L pots and watered until their reproductive stage. In June 2014, between 9 and 16 F1 plants per family were self-pollinated (using several flowers of each plant) to obtain a total of 2000 (F2) full-sib seeds. We used 25 seeds/family to produce a first batch of plants that were maintained within the shadehouse in the facilities of CICOLMA under homogeneous environmental conditions, restricting their contact with herbivores, pollinators, and patrolling ants. A second batch of plants of

the same genetic families was produced using 75 seeds/family to expose them to all biotic interactions and natural environmental variation in the field. Seeds were germinated as described above. For the shadehouse batch of F2 plants, we assigned 20 seedlings per family to one of the following three ontogenetic stages: *Seedlings* (individuals with the first two true leaves fully expanded), *Juvenile* (plants with the 10th leaf fully expanded), and *Reproductive* (individuals with fully expanded leaves bearing the first flowers). We transplanted plants to individual germination trays (seedlings), 1-L pots (juvenile plants), or 2-L pots (reproductive plants), and watered them daily. We did not observe pot binding of roots at any ontogenetic stage, and due to the small root size of seedlings and juvenile plants, it is unlikely that pot size affected plant growth or other traits at any ontogenetic stage. For the field batch, 1200 seedlings were transplanted at the 1st true leaf stage directly into the soil, assigning them to one of the ontogenetic stages in which they were later measured ($N = 20$ plants/family/stage). We established twenty 1×1 m plots in the sand dune under the partial shade of vegetation canopy, where *T. velutina* plants naturally grow. Between 3 and 10 plants/family were planted in each plot using a 10×10 cm grid.

Ontogenetic trajectories in plant defense

For both batches, when plants reached their corresponding ontogenetic stage (i.e., plants were measured only once), we quantified EFN sugar and collected two fully expanded leaves, one to assess cyanogenic potential and one to quantify trichome density.

We estimated cyanogenic potential through HCN content (Ballhorn et al. 2005), which was assessed by a quantitative and colorimetric assay (Schappert and Shore 1995) using the most apical fully expanded leaf at each ontogenetic stage. In each leaf, we cut six leaf discs (0.6 cm^2) with a whole punch. Three discs were stored in a glassine paper bag and dried at room temperature, until constant dry weight was recorded. We crushed the remaining three discs in an Eppendorf tube with $7 \mu\text{L}$ of chloroform. A 0.5×1 cm strip of filter paper previously soaked in a 5% NaCO_2 and 0.5% of picric acid solution was then suspended inside the tube, avoiding direct contact with plant material. We left the tubes in darkness for 24 h at room temperature ($25\text{--}30^\circ\text{C}$). The picrate paper changed from yellow to orange, rust, or dark brown depending on the amount of HCN released by the leaf discs. We punched a single disc from the reacted picrate-soaked filter paper and eluted it in 1 ml of 50% ethanol for 24 h in the fridge at 4°C . $250 \mu\text{L}$ of the eluted ethanol was then placed in microplates (96-well EIA/RIA plates, Corning, NY, USA) to measure absorbance at 590 nm using a microplate reader (ELx808, BioTek Instruments Inc., Winooski,

VT, USA). We transformed absorbance readings to HCN contents ($\mu\text{g HCN/g dry weight}$) using the formula $\text{HCN} (\mu\text{g}) = (\text{optical density} - 0.0478965)/0.000652$ ($r^2 = 0.91$, $P < 0.0001$), obtained from a standard curve using the same protocol as described above, but using sodium cyanide (Code 7660-1 Caledon Laboratories Ltd, Canada) as a source of HNC.

To quantify trichome density, we used the penultimate apical fully expanded leaf of each plant, after leaves were dried at room temperature, and no changes in weight were observed. Two pictures were taken of the upper and lower sides of each leaf using a stereoscopic microscope (Discovery V8, Zeiss, $\times 1.5$). We quantified the number of trichomes per area using the software Image J 1.48v (NIH, USA). We estimated the production of sugar in EFN in the three most apical leaves following the procedures described by Heil et al. (2001). Briefly, we added $2 \mu\text{L}$ of distilled water to the extrafloral nectar using a micropipette ($0.5\text{--}10 \mu\text{L}$ Nichipet Premium, Nichiro CO, Japan), without completely discharging the volume of water into the nectary. Then, the mix of water and nectar was reabsorbed and placed in a hand-held refractometer ($0\text{--}50^\circ$ Brix, Reichert 137531L0, Munich, Germany) to quantify sugar concentration. Finally, the mix of nectar and water was reabsorbed again from the refractometer using $5\text{-}\mu\text{L}$ capillary tubes (Blaubrand intra-MARK, Brand, Germany) and its volume was estimated by measuring the length of the nectar column with a caliper. Sugar content in EFN was estimated as $\text{sugar (mg)} = \text{nectar concentration (} B^\circ \text{)} \times \text{nectar mix volume (}\mu\text{L)} / 100$ (Heil et al. 2000, 2001).

Risk of attack and herbivore damage

To assess the selective potential of herbivores on different ontogenetic stages of *T. velutina*, we estimated butterfly oviposition preferences of adult stages of *E. hegesia*, for different *T. velutina* ontogenetic stages. In addition, we quantified the percentage of leaf area consumed by herbivores, as a measure of the potential impacts on plant fitness and hence the intensity of selection pressure on defensive traits. To assess oviposition preference of *E. hegesia* butterflies on different ontogenetic stages of *T. velutina*, we first reared caterpillars on *T. velutina* plants until pupation. Once they emerged, 11 pairs of butterflies (one female and one male) were placed into one individual cylindrical cage made of a fine mesh net and a wire circle ($N = 11$ cages of 60 cm of diameter $\times 100 \text{ cm}$ in height). Within each cage, we placed four 2-L pots, two of them containing a single reproductive plant and two containing a group of ten seedlings to match the number of leaves of reproductive plants. A cotton ball dipped in artificial nectar (Schappert and Shore 1998) was hung from the top of the cage to provide food for the butterflies, in addition to flowering *Eupatorium* sp. and *Lantana*

camara cuttings. Only five pairs of butterflies succeeded to mate. After mating, females were kept inside the cages for 48 h to allow oviposition. We counted the number of eggs on seedlings and reproductive plants. Oviposition preference was estimated as the number of eggs/total leaf area available of each ontogenetic stage within each cage (i.e., adding the leaf area of all individuals/stage). After verifying the normality of the data, differences between ontogenetic stages were assessed using a paired *t* test (R Core Team 2017).

For a subset of the F2 plants growing in the field ($N = 368$), we estimated the percentage of area consumed by herbivores establishing visual categories of damage to all leaves (1 = 0%, 2 = 0–25%, 3 = 25–75% and 4 = 75–100%). Damage was estimated in each plant at the seedling, juvenile and reproductive stage. Ontogenetic differences in the percentage of leaf area lost were assessed using a repeated-measures linear mixed model with restricted maximum likelihood (REML) estimations, considering family as a random factor and ontogenetic stage as a fixed factor. Significance of the fixed factor was estimated using ANOVA type III SS. We did not test for the significance of random factors, as we were only interested in testing the ontogenetic differences in herbivore damage.

Genetic variance, heritability, and evolvability of plant defenses and their ontogenetic trajectories

To assess if ontogenetic trajectories of defensive traits had the pre-requisites to evolve by natural selection, we assessed their genetic variance, broad sense heritability (hereafter heritability), and evolvability. As a first step and to understand the possible constraints on the evolutionary potential of these trajectories, these attributes were estimated for each trait within each ontogenetic stage and environment, using the variance components obtained through linear mixed-model ANOVAs with restricted maximum likelihood (REML) estimations. Models included each trait as the response variable and family as a random factor. Genetic variance was determined repeating the models with and without the family term and using likelihood ratio tests for significance (Littell et al. 1996; Zuur et al. 2009). Full-sib broad sense heritabilities (H^2) were obtained by dividing two times the variance component of the family term by the total variance (Roff 1997), and significance was tested using the *P* value of the genetic family term (Bingham and Agrawal 2010). For comparative purposes with other studies, evolvabilities for each defensive trait were calculated at each ontogenetic stage as $CV = (\text{variance} \times 2)^{1/2}/\text{mean trait value}$ from models including only family as a random factor (Bingham and Agrawal 2010). To explore the relationships among all defensive traits within and across ontogenetic stages, Pearson correlations were calculated from family means for each ontogenetic stage and defensive trait.

To estimate genetic variation, heritability, and evolvability of the ontogenetic trajectories of the three defensive traits, we used the same approach as in Bingham and Agrawal (2010), in which they estimated heritability and evolvability on the inducibility of defensive traits (i.e., plasticity across different herbivore-damage environments). In our case, we treat ontogenetic stages as different “environments”, and estimate variation, heritability, and evolvability of the change in defensive traits across ontogeny. Because ontogenetic trajectories among the three stages were not linear, we had to partition our data to compare different fractions of the whole ontogenetic trajectory: seedling vs. juvenile and juvenile vs. reproductive. For each of the two data subsets, we used mixed models ANOVAs including ontogenetic stage as a fixed factor, family as a random factor, and their interaction, using log-transformed data. We visually inspected the residual plots to check for deviations from homoscedasticity and normality. To test for significance of random factors, we used the likelihood ratio statistic, taking the difference between the -2REML log likelihood of the full model (including ontogenetic stage, family and their interaction) and a reduced model (without the random factor and/or the interaction; Littell et al. 1996; Zuur et al. 2009). Genetic variation in the ontogenetic trajectories was inferred from the significance of the ontogenetic stage $\times$ family term. To assess H^2 of the trajectories between each pair of ontogenetic stages, we used two times the family $\times$ ontogenetic stage variance component, divided by the total phenotypic variance (Relyea 2005; Bingham and Agrawal 2010). The significance of the heritability was tested using the *P* value of the interaction term.

We calculated evolvabilities of the ontogenetic trajectories between each pair of ontogenetic stages as $CV_G = (\text{variance} \times 2)^{1/2}/(\text{mean trait value of stage } i - \text{mean trait value of stage } j)$, where *i* and *j* correspond to the younger and older stage, respectively, and the *variance* term was obtained from the family $\times$ ontogenetic stage component of the full model used to calculate H^2 of the trajectories. Absolute values of evolvability were used to avoid negative values of decreasing ontogenetic trajectories, and were included for comparative purposes with other studies. We interpret all these genetic effects with caution because plant material was obtained from a single population and may not reflect broad patterns for *T. velutina*.

Developmental reaction norms

To estimate plasticity of the ontogenetic trajectories of each defensive trait, we performed full factorial linear mixed ANOVA models with REML, using ontogenetic stage and environment as fixed factors, and family with all possible interactions as random factors. Significance of ontogenetic stage $\times$ environment term was considered as evidence of

developmental plasticity (Pigliucci and Schlichting 1995). Significance of random factors was tested as described above. All statistical analyses were performed in R 3.3.0 (R Development Core Team 2017), using lme4 package (Bates et al. 2015).

Results

Ontogenetic patterns in plant defense and risk of herbivore attack

As previously reported, we found non-linear ontogenetic changes in the three focal defensive traits of *T. velutina* (Ochoa-López et al. 2015), showing that the cyanogenic potential decreased and trichome density and sugar in EFN increased during plant development (Table 1, Fig. 2, suppl. material S1). Nevertheless, mean values of these traits and their degree of variation among families were contingent upon the ontogenetic stage and the environment: whereas trichome density and cyanogenic potential had greater expression in the field, plants produced more sugar in EFN within the shadehouse. In addition, greater variation among genetic families in the three defensive traits was observed in the field than within the shadehouse, but as a function of plant ontogeny: seedlings showed greater variation in cyanide potential; juvenile plants had more phenotypic variation in the expression of trichome density; and reproductive plants had greater variation in the production of extrafloral nectar than in the other defensive traits (Fig. 2). Interestingly, we found only one significant genetic correlation between trichome density and HCN at the juvenile stage ($r^2 = -0.57$, $P < 0.05$), and only for plants growing in the field. This suggests a possible trade-off between trichomes and HCN at this stage.

We found that *E. hegesia* butterflies preferred to oviposit more frequently on seedlings than on adult plants ($t = 3.308$, $d.f. = 4$, $P = 0.0297$; Fig. 3a). In addition, the percentage of leaf damaged by herbivores was greater for seedlings and decreased across plant ontogeny ($F_{2,736} = 24.42$, $P < 0.001$) such that seedlings had 1.5 and 2 times more damage than juvenile and reproductive plants, respectively (Fig. 3b).

Genetic variation, heritabilities, and evolvabilities of defensive traits and their ontogenetic trajectories

We detected evidence for genetic variation, heritability, and evolvability for trichome density and cyanogenic potential, but only when plants grew in the shadehouse. Moreover, whereas trichome density showed significant heritability and evolvability at all ontogenetic stages, cyanogenic potential was heritable only at the seedling stage (Table 2). In addition, we found that the ontogenetic trajectories of trichome density from seedlings to juvenile and from juvenile to

Table 1 Factorial mixed-model ANOVA testing for the effects of ontogenetic stage and full-sib family on cyanogenesis, trichome density and extrafloral nectar

Environment	Defense attribute	df	F or χ^2	P value
Field plots	Cyanogenic potential			
	Ontogenetic stage	2433	65.15	< 0.0001
	Family	1	1.94	0.1632
	Stage $\times$ family	1	< 0.0001	0.9999
	Trichomes density			
	Ontogenetic stage	2433	101.27	< 0.0001
	Family	1	< 0.0001	0.9999
	Stage $\times$ family	1	< 0.0001	0.9999
	Sugar in extrafloral nectar			
Shadehouse	Ontogenetic stage	1433	60.13	< 0.0001
	Family	1	< 0.0001	0.9999
	Stage $\times$ family	1	< 0.0001	0.9999
	Cyanogenic potential			
	Ontogenetic stage	2962	75.11	< 0.0001
	Family	1	2.34	0.126
	Stage $\times$ family	1	0.82	0.3634
	Trichomes density			
	Ontogenetic stage	2962	245.91	< 0.0001
	Family	1	69.05	< 0.0001
	Stage $\times$ family	1	33.96	< 0.0001
	Extrafloral nectar production			
	Ontogenetic stage	1681	91.06	< 0.0001
	Family	1	< 0.0001	0.9999
	Stage $\times$ family	1	< 0.0001	0.9999

The interaction term shows significant genetic variation of the ontogenetic trajectories and was used to assess the significance of their heritability (Bold values)

*For random effects, the *F*-ratio is replaced with Chi-square values obtained from the difference between the $-2 \log$ likelihoods with that factor included vs. excluded from the model (Littell et al. 1996; Bingham and Agrawal 2010)

reproductive stages showed significant genetic variation (i.e., ontogenetic stage $\times$ family interaction) and significant, although rather low, heritability and evolvability values (Table 3).

Developmental reaction norms

We found significant phenotypic plasticity of ontogenetic trajectories for the three defensive traits, with different trajectories observed in the field vs. shadehouse (ontogenetic stage $\times$ environment interaction, Table 4). However, the developmental reaction norms were different for each defensive trait. For example, cyanogenic potential had contrasting ontogenetic trajectories, with field plants showing significant declines across ontogeny while shadehouse plants tended to show no ontogenetic shifts or only a slight decrease

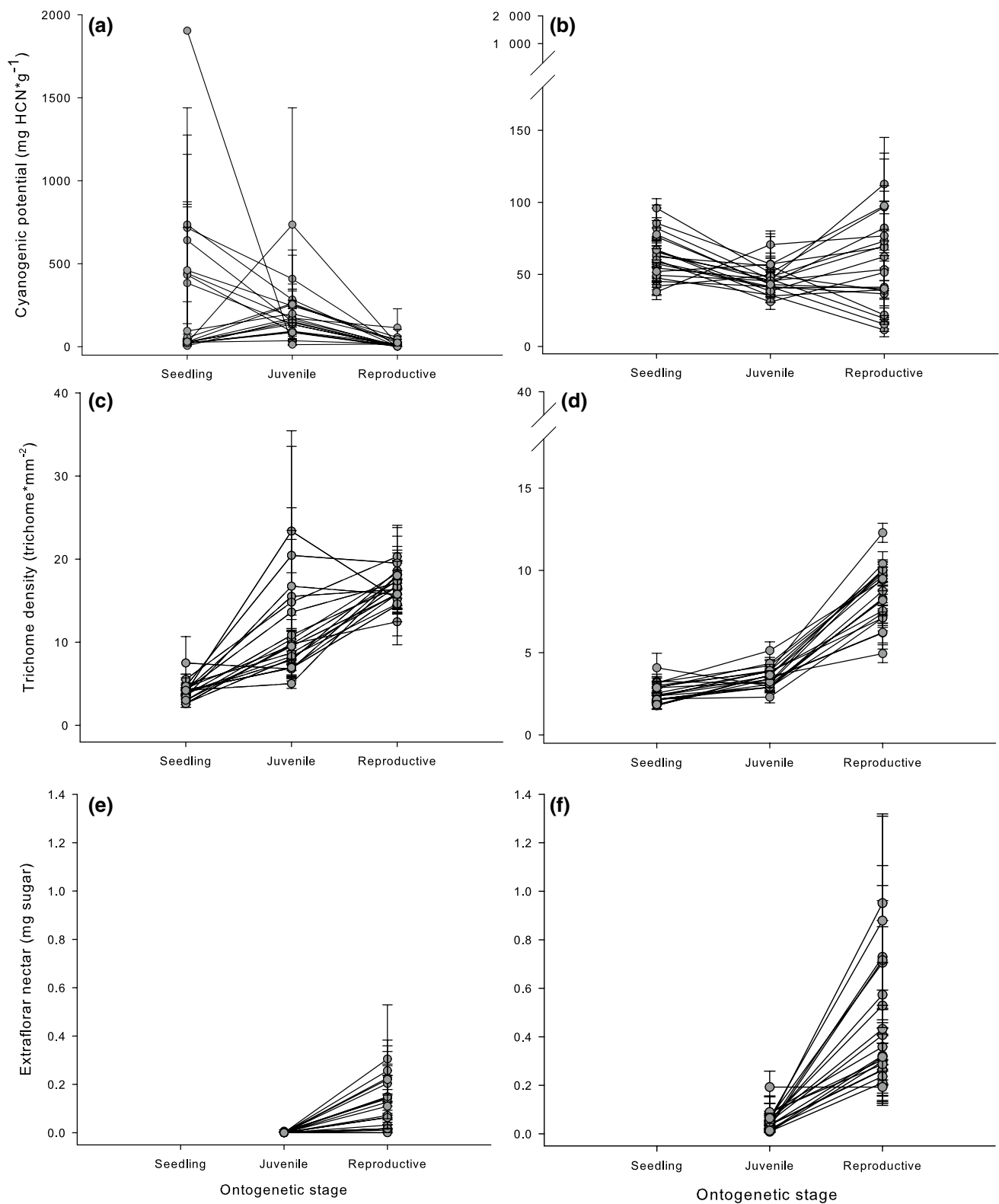


Fig. 2 Ontogenetic trajectories of defensive attributes in 20 genetic families of *Turnera velutina* plants growing in field plots (left side) or within a shadehouse (right side). Lines represent family means with standard error bars. Note the differences in scale

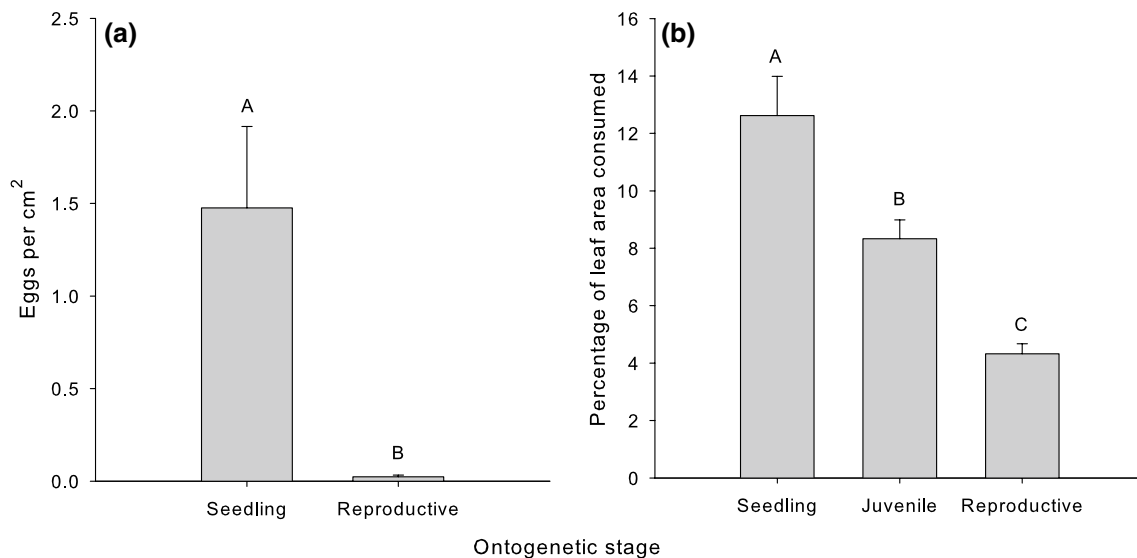


Fig. 3 Risk of attack of *T. velutina* across ontogenetic stages. **a** Oviposition preference of *E. hegesia* butterflies for seedlings or reproductive plants. Different letters represent significant difference among ontogenetic stages. **b** Percentage of leaf area consumed by herbivores

(Fig. 4a). In contrast, trajectories in trichome density and EFN sugar were consistent in directionality for field and shadehouse plants, but differed in the magnitude of change. Ontogenetic increases in trichome density were of greater magnitude in the field (Fig. 4b), while ontogenetic increases in EFN sugar were of greater magnitude in the shadehouse (Fig. 4c).

Discussion

A recent call has been made to move forward from the description of ontogenetic trajectories in plant defense towards the assessment of the mechanisms behind their evolution and/or regulation (Barton and Boege 2017). A first step to better understand the potential of ontogenetic trajectories to evolve through natural selection is to identify the different sources of their variation. In this study, we provide a first report on the genetic and environmental variation of three defensive traits (cyanogenic potential, trichome density and sugar in EFN), assessed at each ontogenetic stage. Our key findings were that the heritable component of variation increased during plant development for trichome density, but decreased during plant development for cyanogenic potential. In the case of trichomes, the amount of heritable variation increased as the risk of attack and herbivore damage intensity decreased, supporting the assumption that natural selection on this defensive trait is likely to be more intense at early ontogenetic stages. In addition, we detected significant heritability for ontogenetic trajectories, but only for trichome

in seedlings, juvenile and reproductive plants of *Turnera velutina*. Different letters represent significant difference among ontogenetic stages ($P < 0.001$). Error bars indicate the standard error of the mean value

density. However, this potential to evolve was contingent upon the environment in which plants grew, as heritabilities were not detected when plants faced different challenges in the field. These findings suggest that ontogenetic trajectories for trichome density in the studied population of *Turnera velutina* show plastic responses and that environmental heterogeneity can actually constrain their evolution as a complex trait.

Genetic variance and heritability of defensive traits in shadehouse plants

We found that the proportion of genetic variation in defensive traits changed as a function of plant age and type of defense. In the case of biotic defenses, only a couple of studies have reported a significant genetic component in the variance for the number (Wooley et al. 2007) and morphology (Rudgers 2004) of extrafloral nectaries, but not on the actual rewards offered to ants (i.e., sugar in EFN), and thus there are no available studies to which we can compare our results. The quality of EFN has been reported to be strongly affected by both the presence and identity of patrolling ants (Heil et al. 2009; Bixenmann et al. 2011), so it is not surprising that we did not find genetic variation in EFN sugar content in the absence of ants within the shadehouse. This lack of significant genetic variation suggests that in the studied population, EFN sugar is a highly inducible trait influenced by environmental variables. Inclusion of larger populations with different ant densities and greater genetic

Table 2 Genetic (V_{Fam}) and environmental (V_{Env}) components of phenotypic variance, coefficients of genetic (CV_G), and environmental (CV_E) variation, and broad sense heritabilities (H^2) for defensive traits within each ontogenetic stage of *Turnera velutina*

Defense attribute	V_{Fam} (%)	V_{Env} (%)	CV_G	CV_E	$H^2 \pm \text{SEM}$
Cyanogenesis					
Field plots					
Seedling	2.09e^{-14} (7.76e ⁻¹³)	2.69 (100)	5.78e^{-10}	0.0117	0.0009 ± 0.1121
Juvenile	0.0996 (2.7)	3.570 (97.3)	0.0027	0.0167	0.05451 ± 0.0289
Reproductive	0.0834 (3.5)	2.266 (96.4)	0.022	0.1149	0.0711 ± 0.0492
Shadehouse					
Seedling	0.062 (8.8)	0.641 (91.2)	0.0055	0.0179	0.1766 ± 0.0473
Juvenile	0 (0)	0.957 (100)	0	0.0297	0.00023 ± 0.0011
Reproductive	0.0741 (2.1)	3.396 (97.9)	0.0069	0.0467	0.0426 ± 0.0215
Trichomes density					
Field plots					
Seedling	0.0079 (9)	0.0795 (90)	0.03	0.0953	0.1806 ± 0.0093
Juvenile	7.40e^{-19} (1.12e ⁻¹⁶)	0.659 (100)	7.79e^{-11}	0.0735	$1.41\text{e}^{-15} \pm 4.94\text{e}^{-15}$
Reproductive	0 (0)	0.109 (100)	0	0.0278	$2.78\text{e}^{-16} \pm 2.78\text{e}^{-16}$
Shadehouse					
Seedling	0.0132 (10.9)	0.1076 (89.1)	0.0617	0.1763	0.2186 ± 0.0431
Juvenile	0.0155 (14.9)	0.0884 (85.1)	0.0501	0.1197	0.2973 ± 0.0571
Reproductive	0.0393 (28.4)	0.0990 (71.6)	0.0329	0.0522	0.5682 ± 0.0854
Sugar in Extrafloral nectar					
Field plots					
Seedling	NA*	NA*	NA*	NA*	NA*
Juvenile	3.22e^{-19} (1.79e ⁻¹²)	1.79e^{-5} (100)	2.37e^{-6}	10.071	$1.27\text{e}^{-14} \pm 4.37\text{e}^{-14}$
Reproductive	1.81e^{-16} (5.51e ⁻¹³)	3.28e^{-2} (100)	1.49e^{-7}	2.0152	$3.56\text{e}^{-15} \pm 3.56\text{e}^{-15}$
Shadehouse					
Seedling	NA*	NA*	NA*	NA*	NA*
Juvenile	0.0004 (4.3)	0.0100 (95.7)	0.5535	2.5923	0.08668 ± 0.0476
Reproductive	0 (0)	0.162 (100)	0	1.2754	$3.045\text{e}^{-15} \pm 1.18\text{e}^{-14}$

Values in parentheses represent the percentage of total phenotypic variation. Significant H^2 values are shown in bold and were inferred from the significance of the family term

*These values are not available because seedlings do not produce extrafloral nectaries at all

variation could shed light on whether this is a general trend for *T. velutina*.

In the case of cyanogenic potential, genetic variation was detected only early in the ontogeny of plants, which does not support our original hypothesis of a negative relationship between the expression of defense, and the amount of heritable variation. The fact that the main herbivore of *T. velutina*, *E. egesia* is a specialist herbivore capable of sequestering cyanogenic compounds (Schappert and Shore 1999), can help to explain oviposition preferences and why this trait has not been fixed as an effective defense at this vulnerable ontogenetic stage. Tolerance, an alternative mechanism to deal with herbivore damage, actually has been reported to be greater for seedlings than in older stages (Ochoa-López et al. 2015), and is likely to represent an alternative mechanism of *T. velutina* to deal with herbivore damage, given the apparent low effectiveness of cyanogenesis. Previous work with *Eucalyptus polyanthemus* has reported that cyanogenic potential is largely genetically determined with little environmental

effects (particularly of nitrogen availability; Goodger et al. 2004). What is novel from our results is the finding that the genetic component of the variance in cyanogenic potential varies across plant development.

For trichome density, we found that heritability increased as plants developed, which actually supports our hypothesis of lower heritability at the most vulnerable stages. However, the fact that this defensive trait also had their lowest levels of expression at this ontogenetic stage suggests that seedlings either rely on other defensive mechanisms (i.e., cyanogenesis or tolerance; Ochoa-López et al. 2015) or have developmental constraints or trade-offs limiting the amount of trichomes they can produce at this stage.

One study has previously reported a decrease in heritability of trichome density during plant development in *Arabidopsis thaliana* (Mauricio 2005). However, in that study, leaf age was not controlled across plant development, and hence it is difficult to disentangle shifts due to leaf ontogeny from those of whole-plant ontogeny. For *T. velutina* in contrast,

Table 3 Variance of genetic and environment interaction ($V_{\text{Fam} \times \text{Stage}}$) and environmental (V_{Env}) components, coefficients of genetic (CV_G) and environmental (CV_E) variation and broad sense heritability (H^2) of two ontogenetic transitions of defensive traits in *Turnera velutina*

Defense attribute	$V_{\text{fam} \times \text{stage}}$ (%)	V_{Env} (%)	CV_G	CV_E	$H^2 \pm \text{SEM}$
Cyanogenesis					
Field plots					
Seedling–Juvenile	0.0763 (2.2)	3.3063 (97.7)	0.0029	0.0193	0.0451 ± 0.0250
Juvenile–Reproductive	0 (0)	3.052 (95.7)	0	0.0187	$1.01\text{e}^{-16} \pm 4.78\text{e}^{-16}$
Shadehouse					
Seedling–Juvenile	0.0248 (2.9)	0.8216 (97.1)	0.0138	0.0795	0.0569 ± 0.0305
Juvenile–Reproductive	0.0170 (0.79)	2.1020 (98.4)	0.0200	0.2225	0.0164 ± 0.0158
Trichomes density					
Field plots					
Seedling–Juvenile	0 (0)	0.49 (100)	0	0.2567	0.00005 ± 0.0004
Juvenile–Reproductive	0 (0)	0.443 (100)	0	0.0071	$9.32\text{e}^{-16} \pm 3.72\text{e}^{-15}$
Shadehouse					
Seedling–Juvenile	0.0091 (8.2)	0.0970 (87.1)	0.1449	0.4714	0.1647 ± 0.0455
Juvenile–Reproductive	0.0175 (14.5)	0.0935 (77.5)	0.0203	0.0469	0.2938 ± 0.0465
Extrafloral nectar production					
Field plots					
Seedling–Juvenile	NA*	NA*	NA*	NA*	NA*
Juvenile–Reproductive	1.18e^{-17} (9.2e^{-14})	0.0128 (100)	3.68e^{-11}	0.0012	$4.84\text{e}^{-15} \pm 2.32\text{e}^{-14}$
Shadehouse					
Seedling–Juvenile	NA*	NA*	NA*	NA*	NA*
Juvenile–Reproductive	0 (0)	0.0835 (100)	0	0.0443	$7.91\text{e}^{-5} \pm 0.0007$

Values in parentheses represent the percentage of total phenotypic variation. Significant heritability values of the ontogenetic trajectories are shown in bold and were inferred from the significance of the family $\times$ stage interaction term

*These values are not available because seedlings do not produce extrafloral nectaries at all

we controlled for leaf age and show that heritability and evolvability of trichome density increased as plants developed from seedlings to reproductive plants. In addition, the existence of significant heritability at the three ontogenetic stages represents an opportunity for ontogenetic trajectories to evolve (see below). In general, both the heritability and evolvability of trichomes and HCN were low compared to what has been previously reported for these traits (Schappert and Shore 2000; Geber and Griffen 2003; Johnson et al. 2009). This might be explained by the fact that we used plant material from a single population. Further investigations should assess if this is the case in different populations of *T. velutina*, which have a non-continuous distribution of along the coastline (Arbo 2005), with probable reduced gene flow among them.

Sources of variation and evolutionary potential of ontogenetic trajectories in defensive traits within the shadehouse

Interestingly, in the absence of the interaction with herbivores and ants under a homogeneous environment in the shadehouse, the ontogenetic trajectories of each defensive trait included different primary sources of variation. In the

case of cyanogenic potential and EFN sugar, ontogenetic trajectories a large proportion of the variation (between 95 and 100%) was unexplained by plant genotypes. In contrast, for trichome density we found significant evolutionary potential of their ontogenetic trajectories at the two developmental transitions, even though a greater heritability was found for the transition from juvenile to reproductive stages than for the transition from seedling to juvenile stages. Hence, if trichome density represents an effective barrier against herbivory at least in one developmental stage, the ontogenetic trajectories could have the potential to evolve through natural selection. However, as discussed below, environmental effects reduced this evolutionary potential under field conditions.

Influence of the environment on the heritable component of defensive traits and their ontogenetic trajectories

Environmental stress has been predicted to either increase or decrease heritable variation, as a consequence of the release and amplification of phenotypic variation (reviewed by Hoffmann and Merilä 1999; Badyaev 2005). In our study, when the same genotypes used in

Table 4 Factorial mixed-model ANOVA for the effects of ontogenetic stage, environment, and full-sib family on cyanogenesis, trichome density and extrafloral nectar*

Defense attribute	df	F or χ^2	P value
Cyanogenesis			
Ontogenetic stage	2,1396	130.34	< 0.0001
Environment	1,1396	24.32	< 0.0001
Family	1	0.84	0.3592
Stage $\times$ environment	2,1396	15.63	< 0.0001
Stage $\times$ family	1	1.41	0.2349
Environment $\times$ family	1	3.87	0.049
Stage $\times$ environment $\times$ family	1	< 0.00001	0.99999
Trichome density			
Ontogenetic stage	2,1396	367.4	< 0.0001
Environment	1,1396	350.69	< 0.0001
Family	1	34.63	< 0.0001
Stage $\times$ environment	2,1396	12.85	< 0.0001
Stage $\times$ family	1	0.8299	0.3623
Environment $\times$ family	1	0.11	0.7352
Stage $\times$ environment $\times$ family	1	8.15	0.004
Extrafloral nectar			
Ontogenetic stage	1,1025	86.84	< 0.0001
Environment	1,1025	38.01	< 0.0001
Family	1	< 0.00001	1
Stage $\times$ environment	1,1025	11.65	0.0006
Stage $\times$ family	1	< 0.00001	0.99999
Environment $\times$ family	1	< 0.00001	0.99999
Stage $\times$ environment $\times$ family	1	0.01	0.9182

Stage $\times$ environment $\times$ family interaction term shows significant genetic variation of the ontogenetic reaction norms of defensive attributes. Bold values indicate significant terms

*For random effects, the *F*-ratio is replaced with Chi-square values obtained from the difference between the $-2 \log$ likelihoods with that factor included vs. excluded from the model (Littell et al. 1996; Bingham and Agrawal 2010)

the shadehouse were exposed to field conditions, the heritable component of cyanogenic potential, trichome density, and the ontogenetic trajectory of the latter were minimized. We suggest that stressful conditions in the field (i.e., herbivore attack, high light intensity and water stress) triggered plastic responses, thereby masking the heritable variation detected in shadehouse plants growing under more homogeneous conditions and in the absence of multiple biotic interactions. Hence, the evolutionary potential of these defensive traits and their ontogenetic trajectories (in the case of trichome density) are likely to be constrained by phenotypic plasticity (Blum 1988; Gebhardt-Henrich and Van Noordwijk 1991).

Developmental reaction norms or plasticity in the ontogenetic trajectories

More than 20 years ago, Pigliucci and Schlichting (1995) proposed the concept of developmental reaction norms to address changes in phenotypic plasticity during ontogeny, and highlighted the importance to merge the study of environmental effects and developmental changes on the expression of phenotypes to understand the mechanisms behind phenotypic plasticity. Ontogenetic reaction norms have been previously reported in plants for phenological timing, plant height, and leaf number (Pigliucci and Schlichting 1995), leaf area and biomass (Turkington 1983; Roach 1986), and also in animals for head morphology in cichlid fishes (Meyer 1987), body length in *Daphnia* (Ebert et al. 1993), and the induction of spines in dragonfly larvae (Arnqvist and Johansson 1998). In this study, we report for the first time developmental reaction norms, or ontogenetic changes in the phenotypic plasticity of three plant resistance traits. In the case of cyanogenic potential and trichome density, both mean values and phenotypic variation were greater in the field than within shadehouse conditions, and the opposite pattern was observed for sugar in extrafloral nectar. This plasticity was probably triggered by the presence and variation of different environmental challenges, such as light exposure, water stress, herbivory, and ant patrolling. For example, the absence of ants within the shadehouse likely triggered an decrease in the rewards to recruit potential defenders.

We also report that ontogenetic reaction norms of trichome density had a genetic component (i.e., significant ontogenetic stage $\times$ environment $\times$ family), indicating that genotypes differed in their ability to modify their ontogenetic trajectories as a function of the environmental challenges. This variation in the ontogenetic trajectories across environments offers the opportunity to further investigate the conditions under which particular trajectories are adaptive, and further identify likely agents of selection and/or developmental constraints. The understanding of the regulatory genes underlying the described ontogenetic changes in these traits would be particularly revealing and enhance our understanding of the mechanisms driving adaptive plasticity (Arnqvist and Johansson 1998; Mauricio 2005). Overall, we provide evidence that the ontogenetic trajectories of some defensive traits are heritable, but contingent upon the environment in which plants grow. From this, we can now evaluate the effect of selective pressures and demographic consequences of these complex phenotypes, to comprehend the mechanisms behind the drivers of evolution in plant defense. Nevertheless, because heritability values reported in this work represent the variation from a single population of *T. velutina*, we recommend further studies to assess genetic variation in ontogenetic

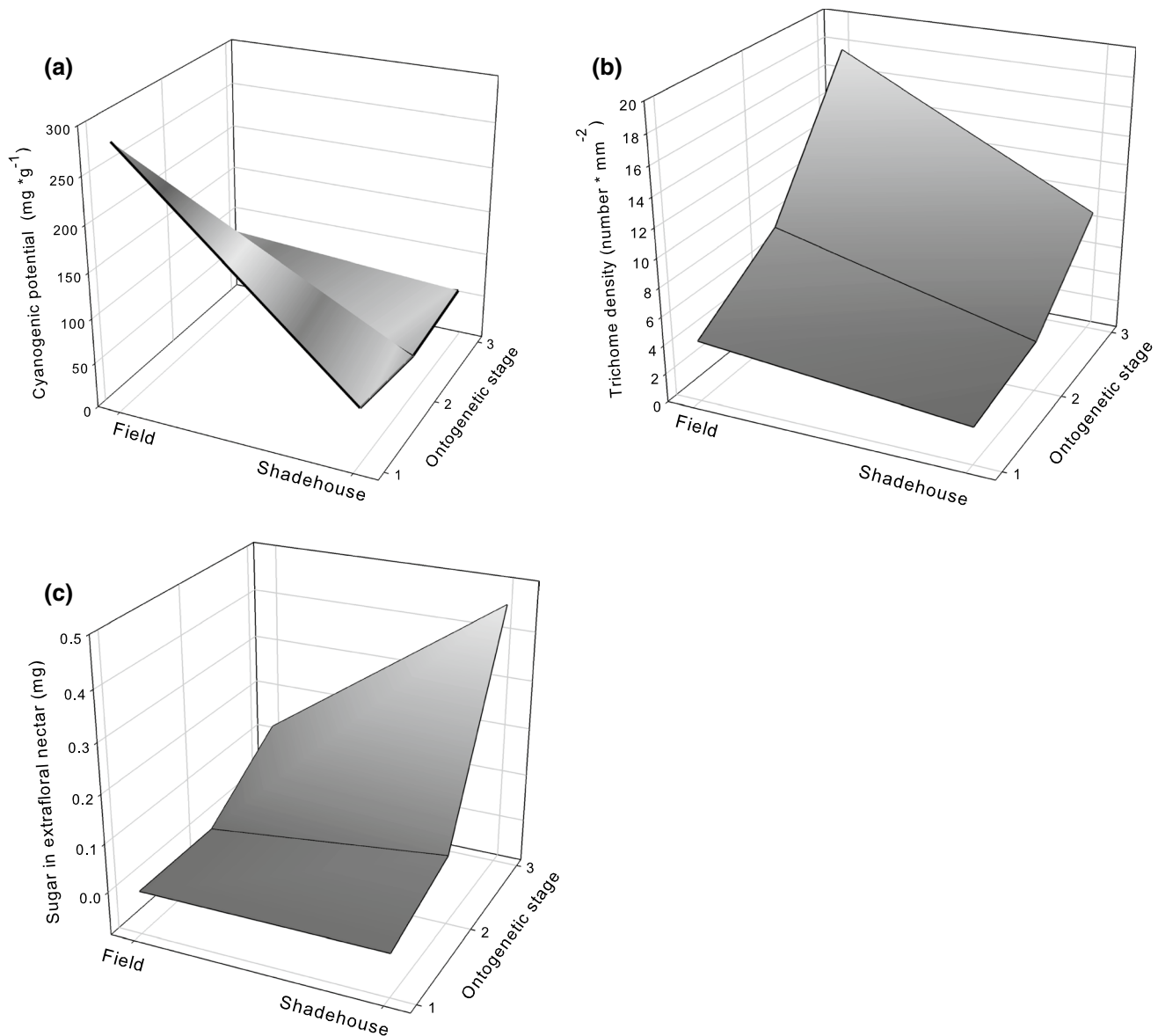


Fig. 4 Ontogenetic reaction norms of *Turnera velutina* of: **a** cyanogenic potential, **b** trichome density and **c** extrafloral nectar expressed during three ontogenetic stages: (1) seedlings, (2) juvenile and (3) reproductive under shadehouse and field conditions

trajectories across populations, to confirm that the trends found for this population represent broad patterns for the species.

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